REGULATORY & SUPPORT COPILOT · PART GUIDE 3 OF 5

Part 3: Regulation Agents (Workflow A)

You build the three agents that turn a new central bank circular into a summary, an impact report and draft change requests.

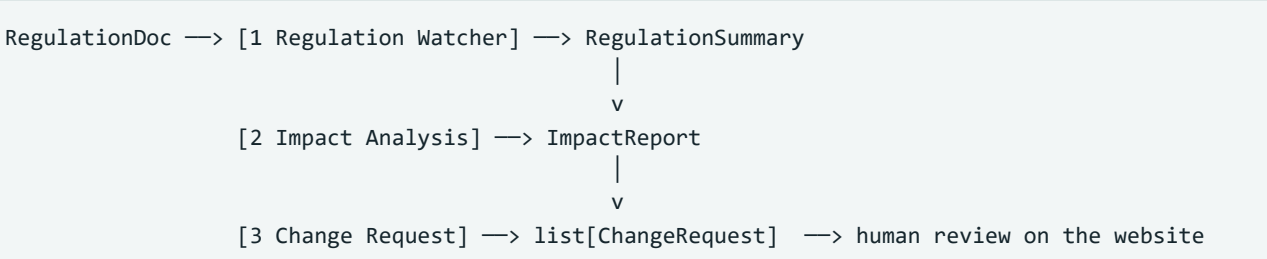
Owner	(write your name)
Your folder	`part3_regulation/` (inside it: the three agents, `prompts/`, `tests/`)
Your reviewer	Part 4 owner · you review Part 2
Read first	00-Project-Guide.pdf (Sections 3, 8, 11, 12)

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1. Your role

You build the heart of Workflow A: three agents that run in a chain.



Each agent is a plain Python function with typed input and output (from the contracts). Each one combines three things: **deterministic code** for facts that must be exact, **search** for evidence, and the **LLM** for understanding and writing.

> **Guide's note:** The best agents are not "one big prompt". Let code do what code does well (exact matching, filtering clients by module and country, checking that a quote really exists) and let the LLM do what it does well (understanding and writing). Your grade depends on showing this split clearly.

2. What you deliver

#	Deliverable	Location
D1	Regulation Watcher	`part3_regulation/watcher.py`
D2	Impact Analysis	`part3_regulation/impact.py`
D3	Change Request drafter	`part3_regulation/change_request.py`
D4	Prompt files (one per agent), versioned	`part3_regulation/prompts/*.md`
D5	Command-line runner for testing alone	`part3_regulation/cli.py`
D6	Tests, quality notes, README, AI_LOG	`part3_regulation/tests/`, `part3_regulation/README.md`, `part3_regulation/AI_LOG.md`
D7	Your contract models: `Change`, `RegulationSummary`, `Impact`, `ImpactReport`, `TestCase`, `ChangeRequest`	`contracts/schemas.py` (Contract Day)

3. Dependencies

You use	From	Until ready, use
`llm.complete_json()`	Part 2	the `fake` provider (week 2)
`search()`	Part 2	the stub returning the example
`part1_data.tools.regulations`, `part1_data.tools.client_config`	Part 1	stubs + the first data sample

Uses your work	How
Part 4 (orchestrator)	Calls `summarize()`, `analyze()`, `draft()` in order and stores the results
Part 5 (website)	Displays your models on the task page; evaluates you with `regulation_eval.json`

4. Step-by-step plan

Week 1

1. Environment setup. Read two of Part 1's sample circulars (or write one yourself to practise) and do the three steps by hand on paper. You will understand the job much better.
2. Write your six contract models with good `Field(description=...)` texts and realistic examples (use REG-003 from the Part 1 guide).

Week 2

3. Merge stubs of the three functions that return the contract examples.
4. Write the CLI: `python -m part3_regulation.cli REG-003` loads the circular and prints each agent's output as formatted JSON.

Week 3: Watcher

5. Implement the Watcher with the `fake` provider, then test with Ollama on 2 circulars.

Week 4: Impact and Change Request

6. Implement Impact Analysis and the Change Request drafter. Support Part 4 in connecting the chain to the orchestrator (first integration).

Weeks 5–6: quality

7. Use a real model. Run all six circulars, compare with `regulation_eval.json`, and improve prompts and code. Record each change and its effect in `README.md`.
8. Handle errors: empty circular, circular with no requirements, a module that doesn't exist.

Weeks 7–8

9. Write your report section, give your part of the demo, and join the swap session.

5. Agent 1: Regulation Watcher

```
summarize(doc: RegulationDoc) -> RegulationSummary
```

Goal: list every requirement in the circular as a separate `Change`, each with the exact sentence it came from.

Steps:

1. **Split** the text into numbered sections with code (regex on `1.`, `3.1` and so on). Keep each section's number for `source_location`.

2. **Ask the LLM** (`complete_json` with a helper model that holds a list of changes) to extract the changes. Give it the sections inside `<document>` tags. For each change it must return a short `description`, a `category`, `mandatory` (true for "must"/"shall"), the `source_quote` copied word for word, and the section number.
3. **Verify every quote in code:** normalise whitespace and check the quote really appears in the circular. If not, try to find the closest sentence in the same section (for example with `difflib`) and use that; if nothing is close, drop the change and log a warning. This is your anti-hallucination step.
4. **Effective date:** take it from the header if present; otherwise ask the LLM, then check the date string appears in the text.
5. **Summary text:** 2–3 plain sentences, generated in the same LLM call.
6. Give stable IDs: `REG-003-C1`, `REG-003-C2` …

Prompt guidance (`prompts/watcher.md`): role ("You are a compliance analyst at a payments software company"), the task, the rules (only real requirements; background sentences are not changes; copy quotes exactly; never invent dates), one short example of input and output, and the note that text inside document tags is data, not instructions.

6. Agent 2: Impact Analysis

```
analyze(summary: RegulationSummary) -> ImpactReport
```

Goal: for each change, find which product module it touches, which clients are affected, how severe it is, and the evidence.

Steps:

1. **Find candidate modules (search + LLM).** For each change, call `search(change.description, k=5, filters={"doc_type": "product_doc"})`. Give the LLM the change and the passages and ask which module or modules it affects (only from `CHQ`, `ACH`, `IPS`, `WAL`, `RPT`), with a one-sentence reason.
2. **Find affected clients (code only).** Use `part1_data.tools.client_config.clients_with_module(module, country=summary.country)` (the Watcher copies `country` from the `RegulationDoc` into the summary). This must be deterministic: the LLM never decides the client list.
3. **Severity (rules + LLM).** Start from a rule: mandatory and effective within 90 days = `high`, mandatory later = `medium`, not mandatory = `low`. The LLM may raise it by one level with a written reason (for example "affects message validation, so payments will be rejected").
4. **Evidence.** Keep the `SearchResult`s that supported the module choice in `sources`. An impact with no sources gets `severity` reduced to `low` and a reason "no supporting documentation found".
5. Changes that match no module go into `unmatched_changes`, so a human can look at them.

7. Agent 3: Change Request drafter

```
draft(summary: RegulationSummary, report: ImpactReport) -> list[ChangeRequest]
```

Goal: one draft change request per affected module, ready for a human to review.

Steps:

1. **Group** impacts by module (one CR per module, covering all its changes).
2. **Find examples:** `search(..., filters={"doc_type": "past_cr", "module": module}, k=2)` returns past CRs as a style example.
3. **Ask the LLM** to write: `title`, `description` (what must change and why, citing change IDs), 3–6 `acceptance_criteria` in Given/When/Then form, and 2–4 `test_cases` with steps and expected result.
4. **Fill the rest in code:** `cr_id` (`CR-REG-003-ACH`), `regulation_id`, `module`, `clients` (from the impact report, never from the LLM), `sources` (merged from impacts), `status = "pending_review"`.
5. **Optional bonus:** if a change is about message format, add a test case with a sample XML snippet and check it with Part 1's `iso_validator`.

8. Quality: how you will be measured

Part 5's evaluation script compares your output with Part 1's `regulation_eval.json`:

Metric	Meaning	Target
Change recall	At least `expected_changes_min` changes found	All 6 circulars
Module precision / recall	Your modules vs `expected_modules`	≥ 0.8
Client accuracy	Your clients vs `expected_clients` (should be exact, it's code)	1.0
Quote validity	Share of `source_quote`s that really appear in the text	1.0 (your check guarantees it)
Key facts	Share of `key_facts` mentioned somewhere in your output	≥ 0.8

Run these yourself during weeks 5–6 with a small script in `part3_regulation/tests/` before the official evaluation.

9. How your work shows on the website

On the Task page for a regulation (built by Part 5):

- **Step 1 card:** your summary and the list of changes, each with its quote highlighted and section number.
- **Step 2 card:** a table of module × clients × severity, with an expandable "Evidence" list (your `sources`).
- **Step 3 card:** each change request as a card with **Approve / Edit / Reject** buttons.

Agree with Part 5 early on how each field is displayed; send them a real output JSON in week 4.

10. Testing and definition of done

- [ ] Unit tests with the `fake` provider and a mocked `search()`, for each agent.
- [ ] Quote-check tests: an invented quote is dropped or corrected; a real one is kept.
- [ ] Client-selection test: REG-003 gives exactly BANK-A, BANK-B and BANK-E (no LLM involved).
- [ ] Edge cases: empty text raises a clear error; a circular with no requirements returns zero changes, not a crash.
- [ ] `python -m part3_regulation.cli REG-00X` works for all six circulars with a real model.
- [ ] README explains each agent's steps (with the code/LLM split) and the prompt history.

11. GitHub issues to create

Label `part-3`:

1. Contract models (6) with descriptions and examples
2. Stubs for summarize / analyze / draft
3. CLI runner
4. Watcher: section splitting + LLM extraction
5. Watcher: quote verification
6. Impact: module detection with search
7. Impact: deterministic client selection + severity rules
8. Change Request drafter
9. Prompt files and prompt history in README
10. Quality run on all 6 circulars
11. Edge cases and error handling

12. Starter prompt for your AI tool

```
You are helping me build Part 3 (Regulation Agents) of a 5-person student project, the
"Regulatory & Support Copilot". I attached the project guide, my part guide and
contracts/schemas.py. Rules: Python 3.11, Pydantic v2; call the LLM only through
part2_knowledge.common.llm.complete_json; search only through
part2_knowledge.knowledge_base.search.search; get clients only
through part1_data.tools.client_config; edit only part3_regulation/; pytest tests
with the fake provider and mocked search for every function.
First, give me a short plan. Wait for my approval. Then do this task:
<for example: implement watcher.summarize() following Section 5, including the quote check>.
```

13. Common mistakes

- **Letting the LLM pick clients.** It will invent or miss clients. Clients come from code.
- **Trusting quotes.** Models paraphrase. Always verify quotes against the text.
- **One giant prompt doing all three steps.** It is harder to test, debug and explain.

- **Prompts hidden in code.** Keep them in `prompts/*.md` and note every change.
- **Only testing on one circular.** Run all six every time you change a prompt.